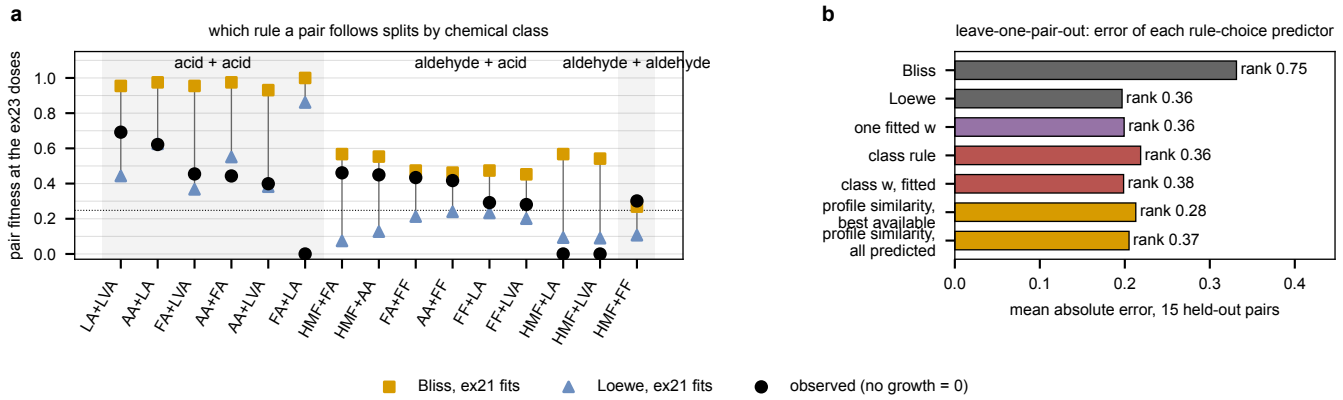



**Figure 12: Which rule a pair follows, and whether anything predicts the choice. a**, Each of the 15 fixed-dose pairs: observed fitness (no growth drawn as zero) against Bliss and Loewe from the titration fits, grouped by chemical class, the dotted line the growth threshold. All 6 acid pairs follow Loewe. The 8 aldehyde and acid pairs split 4 to 4, by the acid and not by the class: with acetic or formic acid the aldehyde follows Bliss, with lactic or levulinic acid it follows Loewe, and 5-HMF with either of those two grew in no well, which Loewe calls and Bliss does not. ✗● **b**, Leave-one-pair-out: each predictor gives a Loewe weight for the held-out pair, fitted on the other pairs, and the pair's fitness is that weight on Loewe plus its complement on Bliss. Mean absolute error over the held-out pairs, with the Spearman over the 12 grown pairs beside each bar. Pure Loewe has the lowest error, 0.197; pure Bliss the best ranking of the grown pairs, 0.748, because its error is a level bias that leaves their order intact. The growth call is not shown: with 12 of 15 pairs grown, Bliss scores 0.800 by predicting growth for every pair, a majority-class readout. A fitted global weight (0.199), the class rule (0.219), a fitted class weight (0.199) and profile similarity with the best available profiles (0.213) or the all-predicted ones (0.205) all land between the two rules. Nothing on file predicts the choice; with 15 pairs that is a small-sample statement. ✗●